

Estimating Factor Loadings with PCA

Mathematical Deep Dive for the GIV Labor Market

GIV Methodology Notes

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Abstract

This document explains how Principal Component Analysis (PCA) is mathematically derived and applied to recover latent factor loadings (λ_i) in the context of the Granular Instrumental Variables (GIV) labor market model. We use the notation from the main presentation to demonstrate the underlying linear algebra, specifically constrained optimization via Lagrange multipliers, eigenvalues, and eigenvectors.

1 The Labor Market Setting

Recall the firm-specific labor demand equation from the presentation:

$$L_{it} = \phi^d w_t + \lambda_i \eta_t + u_{it} \tag{1}$$

Where:

- L_{it} : Variation in labor demand at firm i .
- w_t : Aggregate wage.
- η_t : Unobserved aggregate macro shock.
- λ_i : **Factor loading** (firm i 's sensitivity to the macro shock).
- u_{it} : Idiosyncratic shock to firm i .

The Problem: To construct the optimal GIV ($z_t = \sum \Gamma_i L_{it}$) that satisfies the exclusion restriction, we need weights Γ such that $\sum \Gamma_i \lambda_i = 0$. However, the factor loadings λ_i are unobserved. We must estimate them from the data.

2 Data Preparation: The Residual Matrix

To isolate the macro shock, we assume that the common wage effect $\phi^d w_t$ is either partialled out (e.g., via fixed effects or an initial iterative guess in GMM) or subsumed into the dominant factor. Let Y_{it} be the residual labor variation driven purely by macro and idiosyncratic shocks:

$$Y_{it} = \lambda_i \eta_t + u_{it} \tag{2}$$

Let Y be the $T \times N$ matrix of these observations, where rows are time periods t and columns are firms i . If idiosyncratic shocks u_{it} are sufficiently small and T is large, the variance of Y is dominated by the common macro shock η_t .

3 The Mathematics of PCA

PCA finds the direction (a vector of weights) that captures the maximum variance in the data matrix Y . This direction will represent our best estimate of the factor loadings, λ .

3.1 The Covariance Matrix

Assuming Y is demeaned, the $N \times N$ sample covariance matrix across firms is:

$$\Sigma = \frac{1}{T} Y'Y \quad (3)$$

The entry Σ_{ij} represents the covariance between firm i 's labor demand and firm j 's labor demand over time. Because firms are primarily correlated through the common macro shock η_t , this matrix contains the signal of λ .

3.2 The Optimization Problem

We seek a unit vector $v \in \mathbb{R}^N$ such that the variance of the data projected onto v is maximized. The projection of Y onto v is Yv , and its variance is proportional to:

$$\text{Var}(Yv) = v'\Sigma v \quad (4)$$

We want to maximize $v'\Sigma v$ subject to the constraint that v has a length of 1 ($v'v = 1$). If we don't constrain the length, we could make the variance infinite simply by scaling up v .

3.3 The Lagrangian

To solve this constrained optimization problem, we construct the Lagrangian $\mathcal{L}$. We introduce a Lagrange multiplier, α , for the constraint $v'v - 1 = 0$:

$$\mathcal{L}(v, \alpha) = v'\Sigma v - \alpha(v'v - 1) \quad (5)$$

To find the maximum, we take the derivative of $\mathcal{L}$ with respect to the vector v and set it to zero. Using matrix calculus rules ($\frac{\partial(v'Av)}{\partial v} = 2Av$ for symmetric A):

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial v} &= 2\Sigma v - 2\alpha v = 0 \\ \Sigma v &= \alpha v \end{aligned} \quad (6)$$

3.4 Eigenvalues and Eigenvectors

Equation 6 is the fundamental **Eigenvalue Equation**.

- Any vector v that satisfies this equation is an **eigenvector** of the covariance matrix Σ .
- The scalar α is the corresponding **eigenvalue**.

To see what α represents, pre-multiply Equation 6 by v' :

$$\begin{aligned} v'\Sigma v &= v'(\alpha v) \\ v'\Sigma v &= \alpha(v'v) \\ v'\Sigma v &= \alpha \quad (\text{since } v'v = 1) \end{aligned} \quad (7)$$

This proves that the eigenvalue α is exactly equal to the variance captured by the eigenvector v .

Because our goal was to *maximize* the variance $v'\Sigma v$, we must choose the eigenvector associated with the **largest** eigenvalue. Let's call this principal eigenvector v_1 and its eigenvalue α_1 .

4 Connecting PCA back to the GIV Framework

4.1 Recovering λ_i

In our labor market model, the dominant source of variance across all firms is the macroeconomic shock η_t . Therefore, the principal eigenvector v_1 that captures this maximum variance is our empirical estimate of the factor loadings:

$$\hat{\lambda} = v_1 \quad (8)$$

The specific value $\hat{\lambda}_i$ (the i -th element of v_1) is firm i 's estimated sensitivity to the macroeconomy.

4.2 Recovering the Macro Shock η_t

The unobserved macro shock η_t itself (up to a scaling constant) is simply the data projected onto these loadings, known as the Principal Component Score:

$$\hat{\eta} = Y \hat{\lambda} \quad (9)$$

4.3 Constructing the Optimal GIV

With $\hat{\lambda}$ in hand, we return to the core mechanics of the GIV presentation. We construct the $N \times N$ projection (annihilator) matrix Q that is orthogonal to these factor loadings:

$$\hat{Q} = I - \hat{\lambda}(\hat{\lambda}'\hat{\lambda})^{-1}\hat{\lambda}' \quad (10)$$

We then calculate the optimal weights Γ^* by projecting the firm sizes S :

$$\hat{\Gamma}^* = S' \hat{Q} \quad (11)$$

Finally, the valid Granular Instrumental Variable is constructed:

$$z_t = \sum_{i=1}^N \hat{\Gamma}_i^* L_{it} \quad (12)$$

Because $\hat{\Gamma}^*$ is orthogonal to $\hat{\lambda}$ by construction, z_t is purged of the macro shock η_t , satisfying the exclusion restriction.